Machine Learning

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Agenda

- Learning From Data
- Types of Machine Learning
- Prediction Errors
- Evaluation Metrics
- Overfitting and Underfitting

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Let’s begin the discussion by answering a few questions

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Machine Learning Quiz

What is a fundamental objective when learning from data?

A. To understand what causes noise in the data.

B. To find patterns in the data that are expected to repeat in the future.

C. To make guesses without looking at the data

D. To understand the precise mechanism of data collection.

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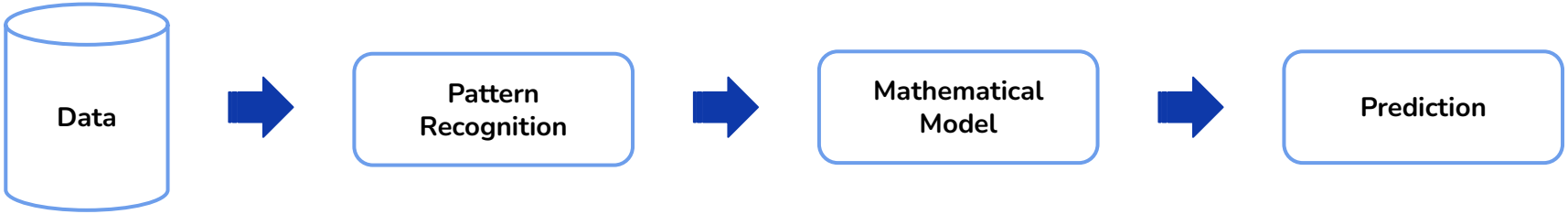
Learning From Data

Data can be messy, with both useful information (patterns) and noise (irrelevant details).

In machine learning, our goal is to find the useful patterns in the data that we can expect to repeat in the future.

To achieve this, we analyze this data and build a model that represents these patterns mathematically.



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Machine Learning Quiz

A company noticed that its website visits increased after previous ad campaigns. Now, it plans to spend $5,000 on ads next month and wants to estimate how many visitors it might get. What kind of approach would best help with this prediction?

A. Regression

B. Classification

C. Clustering

D. Random guessing

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Types of Machine Learning

	Regression	Classification	Clustering
Data Type	Structured data with target variable	Structured data with target variable	Unstructured or structured data without a target variable
Target Variable	Target is numerical	Target is categorical	No target - groups data based on patterns
Goal	Predict a continuous value	Predict a category or class label	Discover hidden patterns or groups in the data
Use Case Example	Use when predicting a number (e.g., visits, sales, prices).	Use when assigning items to categories (e.g., spam vs. not spam).	Use when you want to group similar data without predefined labels (e.g., customer types)

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Machine Learning Quiz

A company uses a linear regression model to predict sales based on advertising spend. How is the error in predictions measured?

A. The slope of the regression line

B. The predicted value

C. The difference between actual and predicted values

D. The division of actual values by predicted values.

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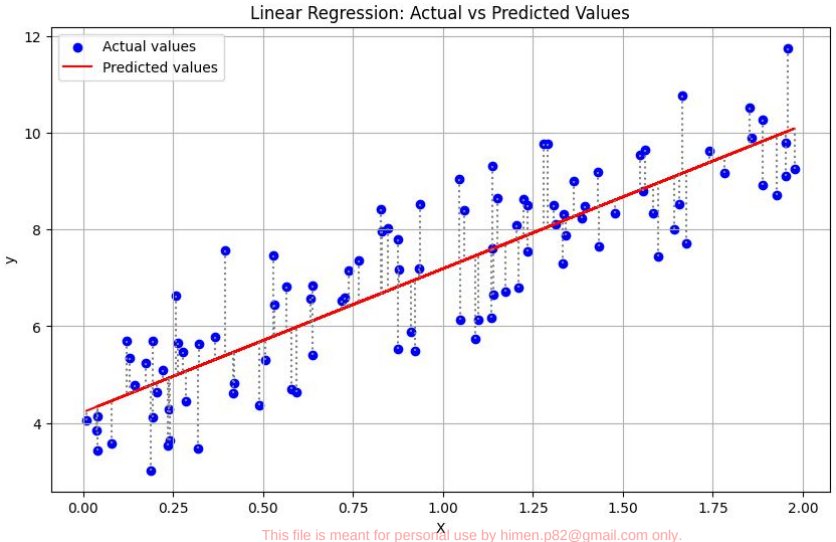
Prediction Errors

In linear regression, an error (also called a residual) is the difference between the actual value and the predicted value of the dependent variable.

Errors help us understand how well the model is fitting the data - smaller errors mean better predictions.



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Machine Learning Quiz

A company used linear regression on past ad spend and visitor data to predict future website visits. After training the model, they now want to evaluate how well it performs, especially by giving more weight to larger prediction errors. Which of the following metrics should they use?

A. Root Mean Squared Error (RMSE)

B. Mean Absolute Error (MAE)

C. Mean Absolute Percentage Error (MAPE)

D. Mean Squared Error (MSE)

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Evaluation Metrics: Regression

Root Mean Square Error	Mean Absolute Error	Mean Square Error	Mean Absolute Percentage Error
• A metric used to measure the accuracy of prediction. • Same unit as the dependent variable • Sensitive to outliers - errors will be magnified due to the square function • Lower the better	• Simplest metric to check prediction accuracy • Same unit as the dependent variable • Not sensitive to outliers i.e. errors don't increase too much if there are outliers • Lower the better	• Penalizes larger errors more by squaring them • Does not use same unit as the dependent variable • Sensitive to outliers due to squared terms • Lower the better.	• Shows average error as a % of true values • Does not use same unit as the dependent variable, but easy to interpret • Can be unstable if actual values are near zero • Lower the better.

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Machine Learning Quiz

A hospital is using an ML model to detect a serious disease. Their top priority is to identify as many diseased patients as possible, even if it means wrongly flagging some healthy patients.
Which evaluation metric is most important in this case?

A. Precision

B. Recall

C. Accuracy

D. F1 Score

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Evaluation Metrics: Classification

Consider a problem of **classifying patients as diseased or healthy**

Four outcomes are possible from an ML model built to do the classification

Case 1	Case 2	Case 3	Case 4
The patient was healthy and the model also predicted that patient would be healthy	The patient was healthy but the model predicted that the patient would be diseased	The patient was diseased and the model also predicted the patient would be diseased	The patient was diseased but the model predicted that the patient would be healthy

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Evaluation Metrics: Classification

Confusion Matrix is a table that helps us summarize those four possible outcomes of the model.

The patient was diseased and the model also predicted the patient would be diseased

The patient was healthy but the model predicted that the patient would be diseased

Predicted Values

Actual Values

	Diseased	Not Diseased
Diseased	**True Positives (TP)**	**False Negatives (FN)**
Not Diseased	**False Positives (FP)**	**True Negatives (TN)**

The patient was diseased but the model predicted that the patient would be healthy

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Evaluation Metrics: Classification

Accuracy: A basic measure used to evaluate the overall performance of a model; represents the ratio of correctly predicted instances to the total instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision: A metric used to measure the accuracy of positive predictions made by a model; represents the ability of the model to reduce the number of false positives.

$$Precision = \frac{TP}{TP + FP}$$

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Evaluation Metrics: Classification

Recall: A metric that measures the proportion of actual positives that are correctly identified by the model; represents the ability of the model to reduce the number of false negatives.

$$Recall = \frac{TP}{TP + FN}$$

F1 Score: In some business scenarios, we might need a model that has both low false positives as well as low false negatives. In such cases, F1 score can be used.

$$F1\ Score = 2 \times \frac{Precision\ \times Recall}{Precision + Recall}$$

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Machine Learning Quiz

Based on the following model performance, which statement best describes the situation?

Model 1: Training Accuracy: 95%, Test Accuracy: 60%
Model 2: Training Accuracy: 40%, Test Accuracy: 35%

A. Both models are underfitting the data.

B. Both models are overfitting the data.

C. Model 1 is overfitting the data, and Model 2 is underfitting the data.

D. Model 1 is underfitting the data, and Model 2 is overfitting the data.

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Overfitting and Underfitting

Overfitting

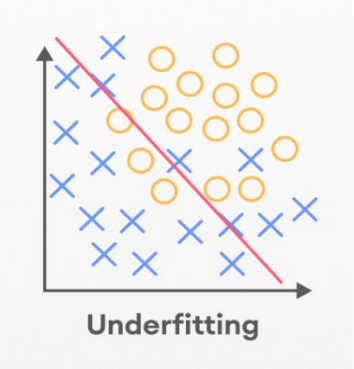
Occurs when models learn noise instead of patterns.

The model performs well on training data but poorly on unseen/test data.

Underfitting

Occurs when the model is overly simplistic.

The model performs poorly on both training and test data.



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Power Ahead!

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